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Preoperative Cryoneurolysis Improves Pain and Function for at Least 12 Months after Total Knee Arthroplasty: A Multicenter Registry Study

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ABSTRACT

Background: With the incidence of total knee arthroplasty (TKA) projected to surpass 1.5 million cases annually in the United States by 2050, optimizing perioperative pain control remains a critical priority. This study investigated the effects of preoperative cryoneurolysis in patients undergoing primary unilateral TKA, focusing on 1) pain intensities, 2) opioid utilizations, 3) functional recoveries, and 4) sleep disturbances over 12 months postoperatively.

Methods: This observational study utilized data from the Innovations in Genicular Outcomes Research, identifying patients undergoing TKA who did versus those who did not receive cryoneurolysis from September 2021 to December 2024, and following them for 12 months postoperatively. Of 356 patients analyzed, 174 received preoperative cryoneurolysis and 182 served as controls. Baseline demographics were collected or tabulated. Pain level was quantified using the Brief Pain Inventory—Short Form instrument and postoperative opioid use, with sleep disturbance measured using the Patient-Reported Outcomes Measurement Information System questionnaire. Physical function was assessed by the Knee Injury and Osteoarthritis Outcome Score, joint replacement questionnaire. Outcomes were assessed preoperatively and at regular weekly, monthly, and quarterly intervals postoperatively. Comparative analysis was performed using generalized linear mixed-effects regression, with significance set at $P < 0.05$.

Results: Patients who received cryoneurolysis demonstrated significantly lower pain scores (2.0 versus 2.4, $P < 0.001$) and reduced opioid utilization across the year (24 versus 41%, $P < 0.001$). Patients who received cryoneurolysis had improved functional outcomes relative to controls (Knee Injury and Osteoarthritis Outcome Score, joint replacement: 65.2 versus 63.9, $P = 0.004$). Sleep disturbances were also lower in the cryoneurolysis group (52.3 versus 53.3, $P < 0.001$).

One or more of the authors of this paper have disclosed potential or pertinent conflicts of interest, which may include receipt of payment, either direct or indirect, institutional support, or association with an entity in the biomedical field which may be perceived to have potential conflict of interest with this work.

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Conclusions: Cryoneurolysis administered before TKA is associated with reduced long-term opioid dependence, improved pain control, and decreased sleep disturbance for 12 months postoperatively, reaching a minimally clinically important difference. These findings replicate our prior analysis with growing evidence of continued benefit at extended follow-up.
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Total knee arthroplasty (TKA) is among the most commonly performed orthopaedic procedures in the United States, with more than 790,000 cases performed annually [1]. This number is projected to nearly double by 2050 as the population ages and the burden of symptomatic knee osteoarthritis (OA) continues to rise [2]. While TKA is highly effective at restoring function and reducing pain, the postoperative period is often marked by substantial discomfort, which can delay mobilization, prolong hospital stays, and hinder rehabilitation [3]. Inadequate pain control following TKA is also associated with lower patient satisfaction scores, increased risk of complications, and greater healthcare costs [4].

Historically, opioids have been the mainstay of post-TKA analgesia [5]. However, opioid use—even when limited to the immediate perioperative period—is associated with a wide range of adverse effects, including nausea, constipation, sedation, respiratory depression, delayed rehabilitation, and the risk of long-term dependency [6,7]. As such, there has been increasing interest in developing multimodal, opioid-sparing strategies to manage postoperative pain effectively while minimizing reliance on narcotics. Enhanced Recovery After Surgery (ERAS) protocols have emerged in response, incorporating spinal anesthesia, nonsteroidal anti-inflammatory drugs (NSAIDs), acetaminophen, regional anesthesia, and periarticular injections [4,8]. Despite these advances, many patients still require opioids postoperatively, highlighting the need for additional nonopioid analgesic modalities or other modalities that provide durable pain relief.

Cryoneurolysis is one such emerging option [9–12]. This technique uses a handheld probe to deliver sub-zero temperatures to targeted peripheral sensory nerves, inducing reversible Wallerian degeneration of the axons while preserving the nerve's structural framework [13]. The resultant temporary disruption of nociceptive transmission can provide analgesia lasting weeks to months—well beyond the immediate postoperative window—without systemic side effects or the risks associated with opioid therapy [14]. Most commonly, cryoneurolysis is administered preoperatively and targets genicular branches such as the infrapatellar branch of the saphenous nerve and the anterior femoral cutaneous nerve. The procedure is minimally invasive, can be performed in an outpatient setting, and does not interfere with the surgical field or rehabilitation protocols [15].

The aim of this study was to validate our preliminary analyses [14,16], which evaluated the effectiveness of preoperative cryoneurolysis in patients who had no preoperative opioid prescription undergoing unilateral primary TKA by analyzing 1) pain severities, 2) postoperative opioid utilization, 3) functional outcomes, and 4) sleep disturbances over a 12-month follow-up period with a larger sample size. We hypothesized that cryoneurolysis administered before TKA would be associated with improved pain, function, and sleep outcomes at 1-year follow-up relative to patients who did not receive cryoneurolysis.

Methods

Registry

The Innovations in Genicular Outcomes Research (iGOR) registry is a prospective, multicenter observational study (ClinicalTrials.gov ID: NCT05495334) designed to evaluate clinical and health outcomes related to knee OA pain for up to 18 months following treatment [17]. Unlike traditional controlled trials, iGOR is structured to reflect routine clinical practice, with treatment decisions driven solely by shared decision-making between patients and their physicians [18]. The registry captures a wide range of pain management approaches, including oral medications such as NSAIDs and opioids, procedural interventions like radiofrequency ablation and cryoneurolysis, intra-articular injections, and surgical treatments including arthroplasty [15,19].

Patient Cohorts

We identified patients who had a knee OA diagnosis and underwent unilateral primary TKA between September 1, 2021, and December 15, 2024, and who had at least 30 days and up to 12 months of follow-up from the iGOR registry. Only opioid-naïve individuals, defined as patients who had no record of preoperative opioid prescriptions, were eligible. To ensure adequate longitudinal data, patients were required to have a minimum of 30 days of follow-up, with assessments available through 12 months postoperatively. Exclusion criteria included enrollment in any concurrent investigational study that could confound results, planned or prior surgery on a different joint, or incomplete baseline or follow-up data.

Patients were further divided based on whether they did or did not receive cryoneurolysis within 30 days preoperatively (cryoneurolysis versus control patients). The cryoneurolysis group underwent treatment with the Iovera° system (Pacira BioSciences, Parsippany, New Jersey), a handheld device that delivers localized cold therapy to sensory nerves to reduce nociceptive signaling [15,20]. The decision to administer cryoneurolysis was made at the discretion of the treating provider based on routine clinical judgment [15]. All participants provided written informed consent before enrollment in the registry. Following administration of local lidocaine, cryoneurolysis was applied to either superficial nerves (e.g., infrapatellar branch of the saphenous nerve, anterior/lateral femoral cutaneous nerves) or deep genicular nerves (e.g., superior or inferior medial/lateral genicular nerves) using either anatomical landmarks or ultrasound guidance [15]. Patients in the control group underwent TKA without preoperative cryoneurolysis. All patients received standard perioperative care at their respective institutions, with no prescribed opioids the night before surgery or in the preoperative area. Baseline demographic data were collected on age, sex, race/ethnicity, body mass index (BMI), Kellgren-Lawrence grade, American Society of Anesthesiologists (ASA) physical status classification, pain catastrophizing scale, and history of alcohol or substance use.

Pain Control Assessment

There were two measures that were used to assess pain outcomes: 1) the Brief Pain Inventory–Short Form (BPI-SF) [21,22] and 2) a structured questionnaire of postoperative pain medications taken, including opioids. Pain severity was evaluated using a modified version of the BPI-SF, specifically focusing on items three through six. These items assessed patients' perceived pain intensity in four contexts: the worst pain, the least pain, the average pain, and the current pain over a 24-hour period. Each item was scored on a numeric scale from 0 (no pain) to 10 (the most severe pain imaginable). Patients recorded their responses at the time of enrollment and then at regular intervals—weekly for the first month, monthly through 6 months, and quarterly through 12 months postoperatively. Postoperative opioid use was tracked through structured patient self-reporting of any use of hydrocodone/acetaminophen, oxycodone, oxycodone/acetaminophen, tramadol, and others (in free text) at the same frequency as pain severity in BPI-SF.

Functional Assessment

Functional recovery was evaluated using the Knee Injury and Osteoarthritis Outcome Score for Joint Replacement (KOOS, JR), a validated patient-reported outcome tool focused on knee-specific health status [16,23]. The survey includes seven items that assess pain, stiffness, and the ability to perform routine physical activities. Each item is rated on a scale from 0 (no problems) to 4 (extreme problems), yielding a raw score range from 0 to 28. Assessments were completed preoperatively and repeated weekly for the first month after surgery, then monthly through 6 months, and quarterly for up to 12 months. Raw scores were converted to a standardized interval scale ranging from 0 to 100, where higher values indicate better knee function. A score of 50 reflects average function in the reference population, while a score of 100 denotes optimal joint health.

Sleep Disturbance Assessments

Sleep disturbances were assessed using the Patient-Reported Outcomes Measurement Information System (PROMIS) Sleep Disturbance Short Form 8b, a standardized instrument designed to evaluate patients' subjective experiences of sleep over the preceding week [24,25]. The questionnaire consists of eight items addressing aspects such as sleep quality, difficulty falling and staying asleep, and the extent to which sleep felt restorative. Raw scores range from 8 to 40, with higher values indicating greater disturbance. These scores were then converted to T-scores, standardized to a general population mean of 50 with a standard deviation of 10. A T-score of 60 reflects sleep disturbance one standard deviation above the population average, denoting poorer sleep quality.

Data Analyses

Baseline characteristics, including demographic and clinical variables, were summarized using descriptive statistics for both the cryoneurolysis and control groups. Comparisons between cohorts were conducted using *Chi*-square tests for categorical variables and Kruskal–Wallis tests for continuous variables. To assess postoperative trajectories in pain, physical function, and sleep quality over a 12-month period, longitudinal trends were analyzed across all available follow-up time points. Minimal clinically important differences (MCIDs) were defined using the Kazis-effect size approach, which accounts for variability from baseline scores. Multivariable generalized linear mixed-effects models were performed to analyze the primary outcomes of interest: pain severity, opioid consumption, functional status, and sleep disturbance [15]. Continuous outcomes were modeled using a normal distribution, while binary outcomes, such as opioid use and MCID achievement, were analyzed using a binomial distribution. All models were adjusted for relevant baseline covariates, including age, sex, BMI, Kellgren-Lawrence grade, ASA classification, baseline pain catastrophizing scale score, timing of TKA (yes/no at study enrollment), and the respective baseline outcome score. Statistical analyses were performed using Statistical Analysis Software software version 9.4 (SAS Institute, Cary, North Carolina), with significance defined as $P < 0.05$.

Baseline Characteristics

A total of 356 patients with no preoperative opioid intake who underwent unilateral primary TKA were included in the analysis and followed for a mean of 301 days postoperatively (Table 1). Overall mean follow-up was 301 days, with a standard deviation of 167 days. Among these, 174 patients received preoperative cryoneurolysis between three and 30 days before surgery to support postoperative outcome improvement, such as pain reduction. The average age of the cohort was approximately 67 years, with 60% identifying as women. The mean BMI was 30.5 (range, 13 to 52), and most patients presented with advanced radiographic OA, with 94% classified as Kellgren-Lawrence grade 4. In terms of overall health status, 74% of patients were rated ASA I or II, indicating low to moderate surgical risk. Overall, patients reported moderate baseline pain (mean BPI-SF: 4.67), suboptimal joint function (mean KOOS, JR: 49.8), and above-average sleep disturbance (mean PROMIS-SD: 54.4) (Table 1) with no significant differences between the two cohorts ($P > 0.05$ for all).

Results

Enhanced Recovery after Surgery and Length of Facility Stay

A similar distribution of several ERAS components was observed between the two cohorts. The frequency of use in cryoneurolysis compared to control cohorts was 30 versus 29% (perioperative corticosteroids), 57 versus 41% (tranexamic acid), 39 versus 42% (antiemetics), 5 versus 10% (urinary catheter), 15 versus 21% (antithrombotics), and 28 versus 35% (wound drains). Some form of nerve block was also administered to at least 90% of patients in both cohorts, with at least 88% among them receiving an adductor canal block. Commonly applied anesthesia to nerve blocks includes liposomal bupivacaine (12%), conventional bupivacaine, and/or ropivacaine (51%). Regarding facility stays, patients in both cohorts were all same-day discharge, with a mean of 19 hours in the cryoneurolysis cohort and 16 hours in the control cohort, respectively.

Decreased Postoperative BPI-SF Pain Severity Over 12 Months in Cryoneurolysis Cohort

Pain severity was reduced and apparent starting at week one following TKA in both cohorts; the least square mean and standard error for pain severity at week one were 3.77 (0.22) in the cryoneurolysis cohort and 4.13 (0.21) in the control cohort (Figure 1). Over 12 months of follow-up, there was a continued trend toward decreased pain severity in both cohorts ($P < 0.001$; Figure 1). Patients treated with preoperative cryoneurolysis had significantly lower pain severity over time than controls at all time points ($P < 0.001$). Additional analysis of MCIDs showed that overall,

Table 1
Baseline Characteristics in the Study Population.

Variable	Control	Cryoneurolysis	Total	P-Value
	N = 182 (%)	N = 174 (%)	N = 356 (%)	
Follow-up time since TKA (day), mean (SD)	268 (159)	334 (169)	301 (167)	<0.001
Age in years, mean (SD)	65 (8.4)	67 (8.8)	66 (8.6)	0.062
Sex, N (%)				0.682
Women	107 (58.8)	106 (60.9)	213 (59.8)	
Men	75 (41.2)	68 (39.1)	143 (40.2)	
Race/ethnicity, N (%)				0.010
Asian	3 (1.6)	5 (2.9)	8 (2.2)	
Black or African American	5 (2.7)	20 (11.5)	25 (7.0)	
Other/unknown	11 (5.9)	9 (5.1)	20 (5.6)	
White	163 (89.5)	140 (80.5)	303 (85.1)	
BMI, mean (SD)	30.7 (6.22)	30.3 (5.77)	30.5 (6.00)	0.489
KL grade, N (%)				0.002
2 (mild)	5 (2.7)	0 (0.0)	5 (1.4)	
3 (moderate)	14 (7.7)	3 (1.7)	17 (4.8)	
4 (severe)	163 (89.6)	171 (98.3)	334 (93.8)	
History of alcohol or substance abuse, N (%)				0.956
No	179 (98.4)	171 (98.3)	350 (98.3)	
Yes	3 (1.6)	3 (1.7)	6 (1.7)	
ASA classification, N (%)				0.743
I	31 (17.0)	26 (14.9)	57 (16.0)	
II	96 (52.8)	92 (53.5)	188 (53.1)	
III	54 (29.7)	54 (31.4)	108 (30.5)	
IV	1 (0.55)	0 (0.0)	1 (0.28)	
Baseline PCS, mean (SD)	15.9 (10.44)	18.1 (13.3)	17.0 (12.0)	0.808
Baseline pain severity (BPI-SF), mean (SD)	4.3 (2.01)	4.6 (2.3)	4.4 (2.2)	0.353
Baseline physical function (KOOS, JR), mean (SD)	51.4 (14.0)	49.7 (15.8)	50.6 (14.9)	0.551
Baseline SD (PROMIS-SD (8b)), mean (SD)	53.8 (8.1)	53.9 (8.5)	53.8 (8.3)	0.582

TKA, total knee arthroplasty; BMI, body mass index; KL, Kellgren-Lawrence; ASA, American Society of Anesthesiologists; PCS, pain catastrophizing scale; BPI-SF; Brief Pain Inventory–Short Form; KOOS, JR; Knee Injury And Osteoarthritis Outcome Scores; Joint Replacement; PROMIS-SD; Patient-Reported Outcomes Measurement Information System sleep disturbance.

significantly more patients who had cryoneurolysis achieved MCID regarding pain severity than controls over a 12-month follow-up (71 versus 63%, $P < 0.001$) (Supplementary Figure 1).

Decreased Incidence of Postoperative Opioid Use over 12 Months Postoperatively in Cryoneurolysis Cohort

Patients who received cryoneurolysis preoperatively reported using fewer opioids than controls at one week postoperatively (75% in cryoneurolysis versus 94% in the control cohort, $P < 0.001$), which continued to two months (25 versus 45%) postoperatively (Figure 2). There were no differences in opioid use from three to 12 months postoperatively ($P > 0.05$), except for four months ($P = 0.003$). Overall, patients who had cryoneurolysis had significantly lower follow-up opioid exposure than controls (overall average opioid use: 24 versus 41%, $P < 0.001$).

Improved Postoperative KOOS, JR Functional Outcome Over 12 Months in Cryoneurolysis Cohort

Patients in both cohorts experienced functional improvement and physical function beginning at two weeks postoperatively, which continued to improve up to 12 months postoperatively (Figure 3). Overall, patients who received cryoneurolysis had better physical function than controls (overall average: 65.2 versus 63.9, $P = 0.004$). However, MCID analysis revealed a similar proportion of patients in both cohorts who achieved meaningful improvement (overall average: 88.4 versus 87.8%, $P = 0.696$).

Decreased Relative Sleep Disturbance in Patients Receiving Cryoneurolysis

At three months postoperatively, patients demonstrated similar sleep quality relative to the general population. Patients who received cryoneurolysis reported significantly less sleep disturbance than controls (overall average: 52.3 versus 53.3, $P < 0.001$) (Figure 4). In MCID analysis, overall significantly more patients who received cryoneurolysis achieved MCID in sleep disturbance reduction relative to controls (54 versus 47%, $P = 0.003$) (Supplementary Figure 2).

Discussion

This study evaluated the impact of preoperative cryoneurolysis on patients undergoing unilateral primary TKA, with outcomes assessed over up to 12-month follow-up. Specifically, we examined postoperative pain severity, opioid consumption, physical function, and sleep disturbance. Our study with an additional 6-month follow-up and more than four times more patients validated our prior findings by demonstrating that patients treated with cryoneurolysis experienced meaningful and sustained improvements across all four outcome categories [15]. Pain scores were significantly reduced throughout the year, with the greatest divergence occurring in the early postoperative weeks. Patients who received cryoneurolysis were 73% less likely than control patients to use opioids in the months following surgery. Improvements in physical function and sleep disturbance were also observed, with patients who received cryoneurolysis reporting better sleep quality and more consistent functional recovery than patients who did not.

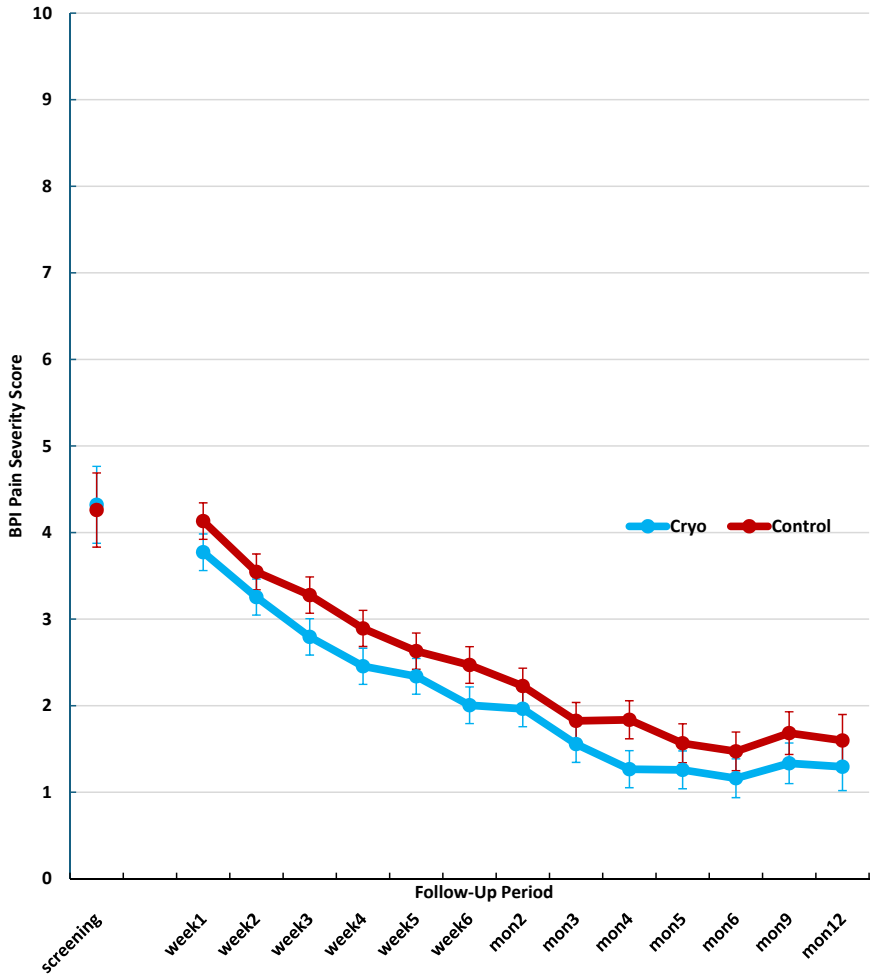

Figure 1. Brief Pain Inventory (BPI) pain severity outcome over 12-month follow-up in cryoneurolysis (Cryo, N = 174) and control (N = 182) groups.

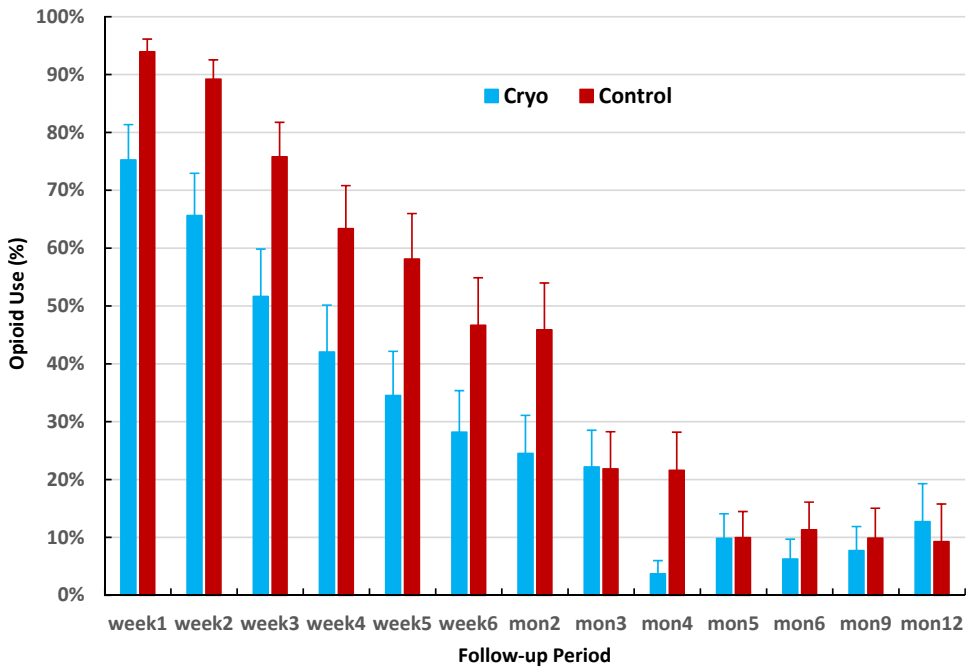

Figure 2. The proportion of self-reported opioid use over 12-month follow-up in cryoneurolysis (Cryo, N = 174) and control (N = 182) groups.

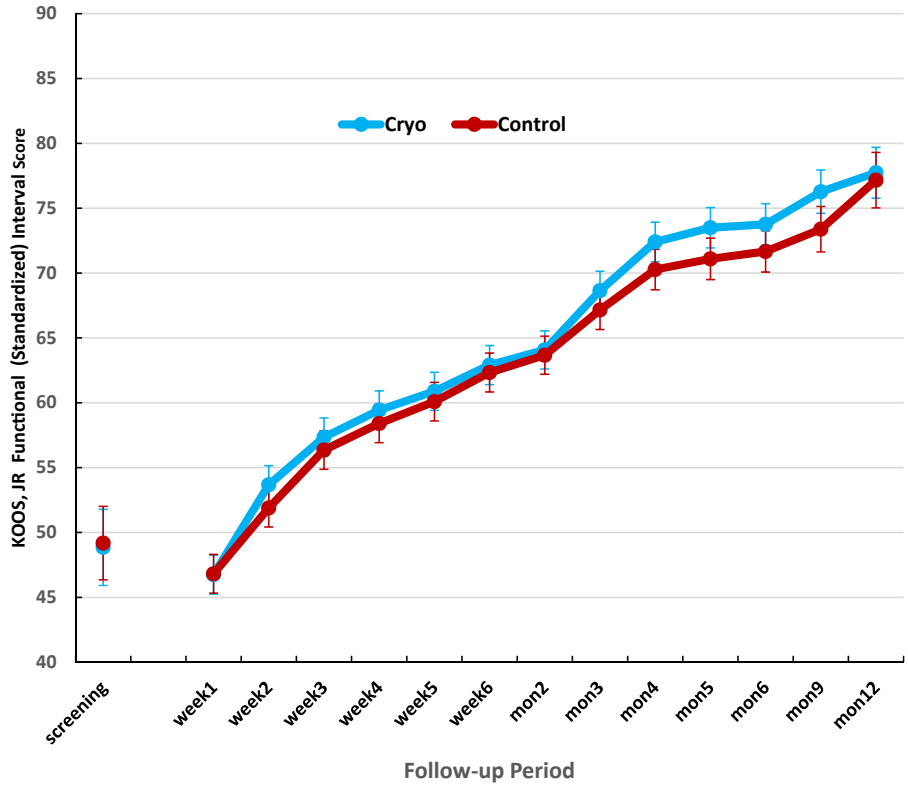

Figure 3. Knee injury and osteoarthritis outcome score, joint replacement (KOOS, JR) functional outcome over 12-month follow-up in cryoneurolysis (Cryo, N = 174) and control (N = 182) groups.

Effective pain management is a cornerstone of recovery following TKA, as it influences ambulation, joint mobility, and patient satisfaction [4]. Long-term opioid use after surgery has been linked to delayed rehabilitation, increased stiffness, and worsened quality of life [7,26]. While enhanced recovery protocols incorporating NSAIDs, acetaminophen, periarticular injections, and neuraxial anesthesia have improved outcomes, these modalities may not provide adequate or durable relief for all patients [27]. Cryoneurolysis presents an alternative or adjunctive strategy, capable of delivering weeks to months of analgesia by disrupting peripheral nociceptive pathways [28]. Cryoneurolysis applies subzero temperatures to targeted sensory nerves, causing reversible axonal degeneration while preserving the nerve's structural integrity [9,13,29]. The technique is minimally invasive, with a low incidence of adverse effects such as skin irritation or transient sensory changes (approximately 1.1%), with none observed in this study [29,30]. Clinical trials and retrospective studies have consistently shown reductions in opioid consumption, improvements in pain control, and faster early mobilization with this approach [13,15,19].

Our findings are also consistent with previous studies reporting favorable pain outcomes associated with cryoneurolysis after TKA. An analysis of 100 patients undergoing TKA by Dasa et al. observed that those receiving cryoneurolysis used 45% fewer opioids over 12 weeks and had shorter hospital stays (6% with stays longer than two days versus 67% in controls, $P < 0.0001$) [31]. Mihalko et al. [32], in a randomized controlled trial, showed that patients who received preoperative cryoneurolysis had significantly reduced opioid consumption at 72 hours, six weeks, and 12 weeks postoperatively (4.2 versus 5.9 mg, $P = 0.0186$). In a retrospective single-center review of 267 patients, Urban et al. [33]found 51% lower daily morphine milligram equivalents (47 versus 97 morphine milligram equivalents, $P < 0.0001$), a 22% lower mean pain score, a 43% reduction in overall length of stay, and greater flexion degree at discharge in the cryoneurolysis versus control group ($P < 0.0001$). Our results are consistent with these findings, demonstrating durable pain improvements and reduced opioid use across a broader patient population with up to 12-month follow-up.

In addition to improved pain control and reduced opioid use, our study suggests that cryoneurolysis may enhance early and sustained functional recovery. Improved KOOS, JR scores were observed in both groups, but patients receiving cryoneurolysis exhibited a more consistent recovery trajectory. This may potentially reflect the downstream benefit of effective analgesia, which facilitates more active participation in physical therapy and promotes overall faster restoration of patient mobility. In a retrospective review of 57 patients undergoing TKA, Lung et al. [34] noted that patients who received cryoneurolysis achieved an average 12-degree increase in knee range of motion and walked 100 feet more than controls at six weeks. These functional gains may translate into improved long-term outcomes when pain and rehabilitation barriers are minimized.

Sleep disturbance is another critical factor influencing postoperative recovery. In our cohort, patients who received cryoneurolysis reported a quicker return to normative sleep quality based on PROMIS sleep scores. This aligns with earlier literature showing that more than 75% of patients who have symptomatic knee or hip OA report sleep disturbances [24,35,36] and that inadequate sleep is linked to worse pain outcomes and slower recovery [25,37]. By mitigating pain and opioid use, both of which can impair sleep, cryoneurolysis may indirectly support more restorative rest and improved recovery trajectories. Of note, as cryoneurolysis causes Wallerian degeneration with direct

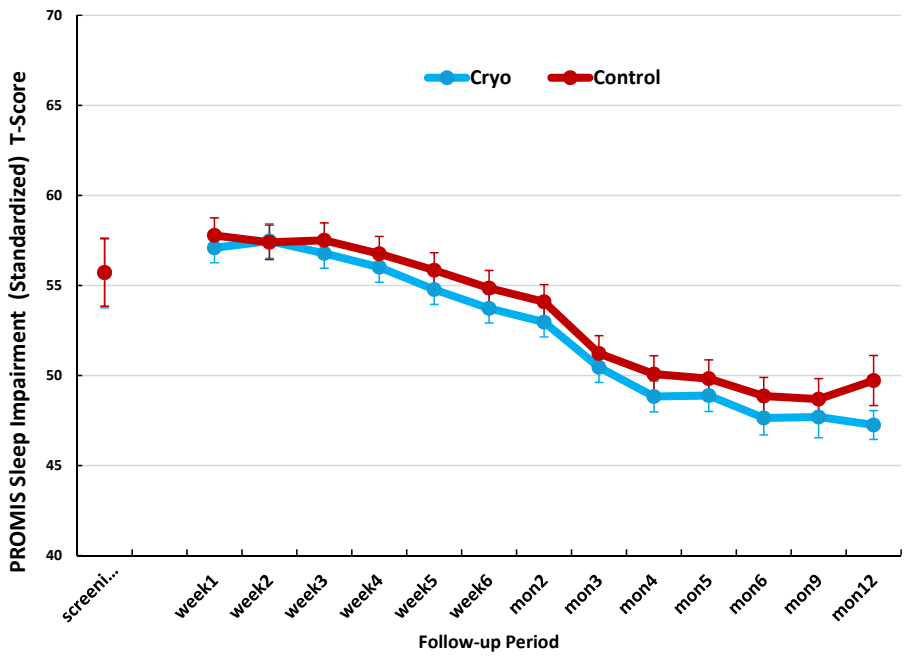

Figure 4. Patient-Reported Outcomes Measurement Information System (PROMIS) sleep disturbance outcome over 12-month follow-up in cryoneurolysis (Cryo, N = 174) and control (N = 182) groups.

analgesic effects up to 12 weeks, sustained differences such as improved sleep may possibly be sustained from indirect effects such as better rehab participation, reduced opioid sleep disruption, and overall decreased pain sensitization.

This study has several potential limitations, most of which are inherent to observational registry-based designs. The absence of randomization or stratification resulted in nonuniform group sizes and variation across sites and subspecialties, which may complicate comparisons between treatment cohorts. Provider-driven treatment selection introduces potential bias, though this reflects real-world clinical decision-making. The database does not allow for identification of the provider administering cryoneurolysis (surgeon versus pain management), and other forms of selection bias based on patients being ambulatory/same-day discharge, geographic differences, and/or site-specific differences regarding cryoneurolysis administration also remain. Specifically, cryoneurolysis adoption likely varied by site based on institutional protocols, provider familiarity, and access, introducing potential selection bias. While adjusted analyses were performed, residual confounding due to site-level differences remains a limitation. While some ERAS components, including tranexamic acid and antithrombotic prophylaxis, were employed in our cohort, their observed utilization rates (tranexamic acid: 41%; deep venous thrombosis prophylaxis: 20%) may appear low for a contemporary TKA population. However, these values reflect real-world documentation variability across sites in the registry, where certain medications (e.g., aspirin or low-molecular-weight heparin) may have been administered but not consistently captured in structured data fields. Additionally, some patients may have received pharmacologic deep venous thrombosis prophylaxis post-discharge in outpatient settings, which were not captured in the intrafacility dataset. Similarly, wound drains were used variably depending on surgeon preference and local protocols, accounting for the 35% usage rate. Overall, the iGOR registry's pragmatic design reflects routine clinical practice rather than protocols, which inherently introduces variability, but improves generalizability. Additional sources of bias include reporting inaccuracies and the Hawthorne effect, where patients may alter their behavior due to awareness of observation. Registry operations were independently managed at each site, which could contribute to variability in data quality based on institutional resources and staffing. Moreover, reliance on smartphone-based data collection may have excluded individuals from lower socioeconomic backgrounds or those who have limited digital literacy, potentially limiting generalizability. The registry also lacks data on treatment costs, technical variations in cryoneurolysis administration, and adverse events, restricting broader analysis of safety and value. Also, while the rate of patients achieving MCID in function was statistically significant, it did not meet thresholds for clinical meaningfulness, underscoring the need for cautious interpretation. Nevertheless, the iGOR registry provides meaningful insight into real-world outcomes across diverse clinical settings and supports the evaluation of emerging treatment strategies in knee OA.

Conclusion

The updated study was conducted to evaluate the impact of administering preoperative cryoneurolysis on opioid-naïve patients undergoing TKA by examining pain levels, opioid use, functional outcome, and sleep disturbance. In summary, patients undergoing TKA who received preoperatively administered cryoneurolysis were associated with better outcomes in pain management and sleep disturbances up to 12 months postoperatively.

CRediT authorship contribution statement

Mitchell K. Ng: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Jennifer H. Lin:** Writing – review & editing, Validation, Supervision,

Software, Resources, Data curation, Conceptualization. **Andrew I. Spitzer:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Funding acquisition, Formal analysis, Data curation. **Vinod Dasa:** Writing – review & editing, Validation, Supervision, Resources, Methodology, Investigation. **Adam Rivadeneyra:** Writing – review & editing, Visualization, Validation, Resources, Project administration, Methodology, Investigation, Funding acquisition. **David Rogenmoser:** Writing – review & editing, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Data curation. **Andrew L. Concoff:** Writing – review & editing, Validation, Supervision, Software, Project administration, Methodology, Investigation. **Mary DiGiorgi:** Writing – review & editing, Visualization, Supervision, Software, Resources, Project administration, Funding acquisition, Formal analysis. **Joshua Urban:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Funding acquisition, Formal analysis, Data curation. **William M. Mihalko:** Writing – review & editing, Visualization, Validation, Resources, Investigation, Funding acquisition, Formal analysis, Data curation. **Michael A. Mont:** Writing – review & editing, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

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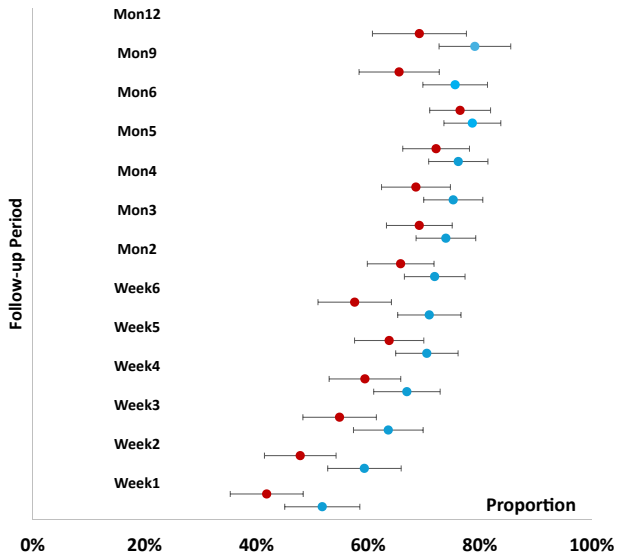
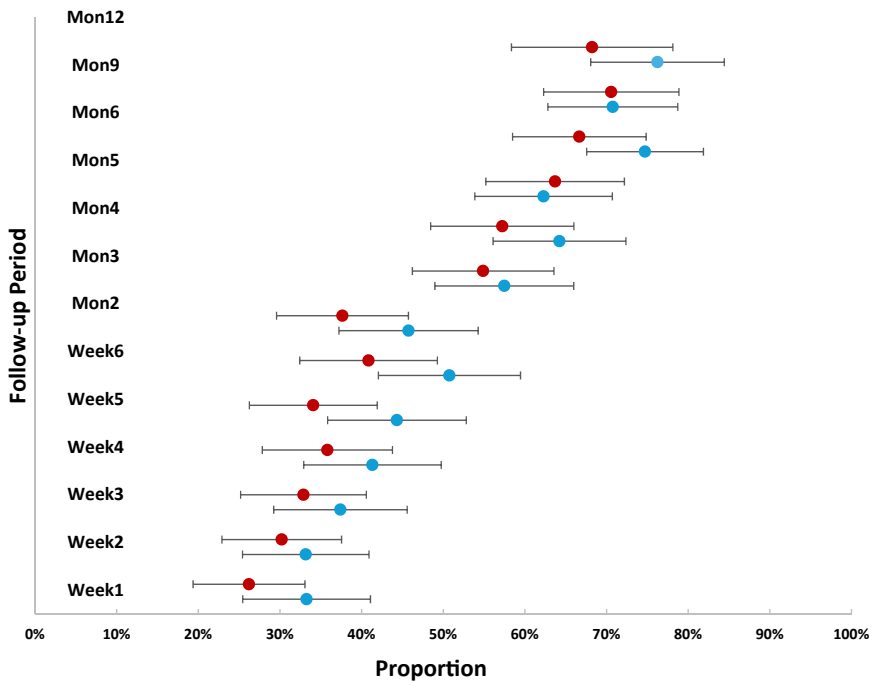
Appendix



Supplementary Figure 1. Proportion (%) and 95% confidence interval of patients achieving minimal clinically important difference (MCID) in pain outcome over 12-month follow-up (blue dot = Cryo; red dot = control).



Supplementary Figure 2. Proportion (%) and 95% confidence interval of patients achieving minimal clinically important difference (mcid) in sleep disturbance over 12-month follow-up (blue dot = Cryo; red dot = control).